

1 Theory

Comparator

A comparator is an electronic circuit designed to compare two input signals and produce an output that indicates which of the inputs is greater or lesser. The output signal reflects the result of this comparison, typically showing a high or low state based on the relative magnitudes of the inputs. Notably, comparators are categorized as non-linear applications of integrated circuits (ICs), as they do not provide proportional amplification but rather switch between discrete output levels.

Truth Table

A	B	$A > B$	$A = B$	$A < B$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

8-Bit Comparator

An 8-bit comparator is a digital circuit that evaluates two 8-bit binary values and determines whether one is larger, smaller, or equal to the other.

2 Implementation and Observation

2.1 8-Bit Comparator

VHDL code - main program

```
library IEEE;
use IEEE.STD LOGIC 1164.ALL;

entity comp is
Port (A: in STD LOGIC VECTOR (7 downto 0);
      B: in STD LOGIC VECTOR (7 downto 0);
      outz: out STD LOGIC VECTOR (2 downto 0));
end comp;

architecture Behavioral of comp is
begin
process(A,B)
begin
if(A > B) then
outz <= "100";
```

```

elsif (A< B) then
outz <= "010";
elsif (A = B) then
outz <= "001";
else
outz <= "000";
end if;
end process;
end Behavioral;

```

VHDL code - testbench file

```

entity comp_tb is
--Port();
end comp_tb;

architecture Behavioral of comp_tb is component comp is port(
    A: in STD LOGIC VECTOR (7 downto 0);
    B: in STD LOGIC VECTOR (7 downto 0);
    outz: out STD LOGIC VECTOR (2 downto 0)
);
end component;

signal A: STD LOGIC VECTOR (7 downto 0);
signal B: STD LOGIC VECTOR (7 downto 0);
signal outz: STD_LOGIC_VECTOR (2 downto 0);

begin
DUT: comp port map( A => A, B => B, outz => outz);
process
begin
A <= "11111111";
B <= "00000000";
wait for 10 ns;

A <= "00000000";
B <= "11111111";
wait for 10 ns;

A <= "00000000";
B <= "00000000";
wait for 10 ns;
wait;
end process;
end Behavioral;

```

RTL Schematic Diagram

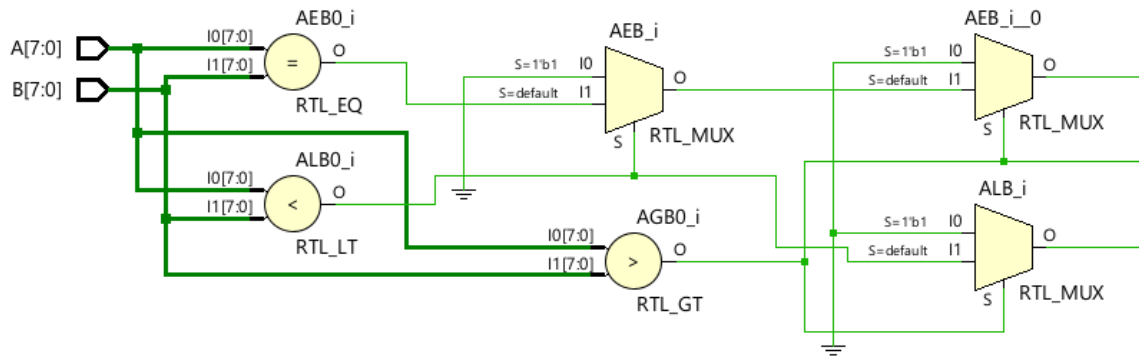


Figure 1: RTL Schematic Diagram of Comparator

RTL design is used to describe the behavior of the comparator as shown in figure 1

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table and code implementation of comparator as shown in figure 2. Here, the output is implemented according to testbench file.

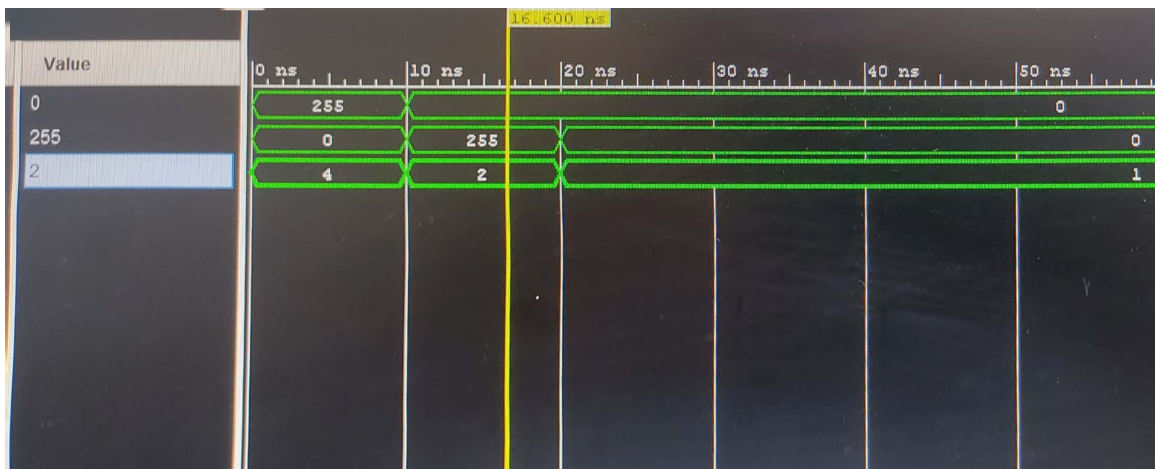


Figure 2: Timing Diagram of Comparator

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 3.

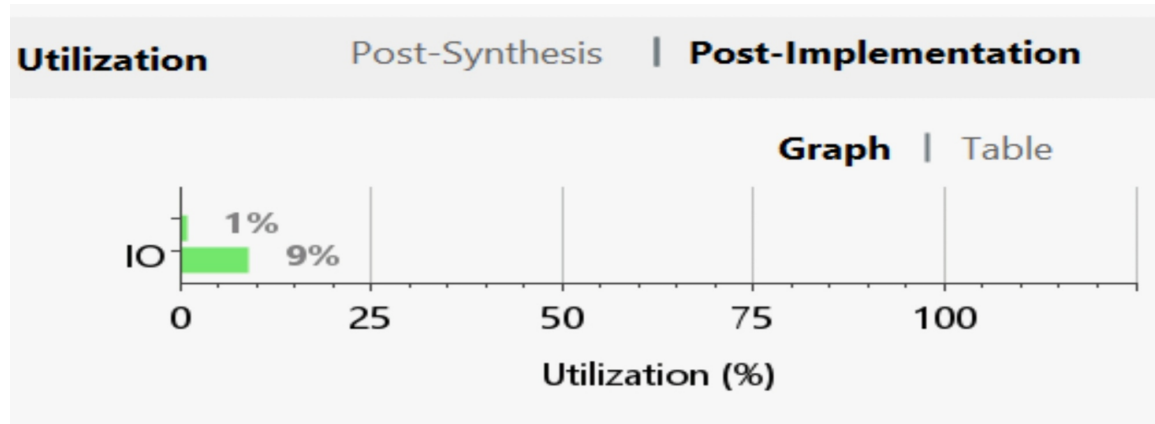


Figure 3: Device Utilization of Comparator

Power Estimation

Figure 4 shows estimation of static power, dynamic power (power required when VHDL program is burned in FPGA) and other related power datas of a comparator circuit.

3 Conclusion

In this lab, we implemented the VHDL code for 8-bit comparator on the Atrix-7 FPGA (part: xc7a50tcsg324-1, package: csg324) using Vivado software. Each design was synthesized, simulated, and implemented, followed by hardware testing, which confirmed their correct functionality.

The process began with simulation to verify the timing diagrams, followed by linting to check for syntax errors. Afterward, the designs were synthesized into a hardware logic circuit, and during the implementation phase, power and timing characteristics were analyzed, displayed and studied. A bitstream file was generated, and the necessary pins were assigned for I/O connections. After successful verification, the design was programmed onto the FPGA board and the desired outputs were obtained according to principles of comparator.

Power		Summary
Total On-Chip Power:		1.143 W
Junction Temperature:		30.5 °C
Thermal Margin:		54.5 °C (11.3 W)
Effective θ_{JA} :		4.8 °C/W
Power supplied to off-chip devices:		0 W
Confidence level:		Low
Implemented Power Report		

Figure 4: Power Estimation of Comparator